

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, c = 0;
6
7     scanf("%d", &n);
8
9     for (int i = n / 2; i < n; i++)
10     {
11         c++;
12         for (int j = 1; j < n; j = 2 * j)
13         {
14             c++;
15             for (int k = 1; k < n; k = k * 2)
16             {
17                 c++;
18                 c++;
19             }
20             c++;
21         }
22         c++;
23     }
24
25     printf("%d", c+2);
26
27     return 0;
28 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

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Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, rev = 0, remainder;
6     int count = 0;
7
8     scanf("%d", &n);
9
10    while (n != 0)
11    {
12        count++;
13        remainder = n % 10;
14        count++;
15
16        rev = rev * 10 + remainder;
17        count++;
18
19        n /= 10;
20        count++;
21    }
22
23    printf("%d", count+3);
24
25    return 0;
26 }
27
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

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Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>  
2 int count=0;  
3 void Factor(int num){  
4     int i;  
5     for(i=1; i<=num; ++i){  
6         count++;  
7         if(num%i==0){  
8             count++;  
9         }  
10        count++;  
11    }  
12    printf("%d", count+1);  
13 }  
14 int main(){  
15     int num;  
16     scanf("%d", &num);  
17     Factor(num);  
18     return 0;  
19 }  
20 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("**");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("**");
                printf("**");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n;
6     int count = 0;
7
8     scanf("%d", &n);
9
10    if (n == 1)
11    {
12        printf("**");
13        count = 2;
14    }
15    else
16    {
17        count = 5 * n + 2;
18    }
19
20    printf("%d", count);
21
22    return 0;
23 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Question 1 | Correct | Mark 1.00 out of 1.00 | [Flag question](#)

Convert the following algorithm into a program and find its time complexity using the counter method.

void function (int n)

```
{
    int i= 1;

    int s =1;

    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n;
6     int i = 1;
7     int s = 1;
8     int c = 0;
9
10    scanf("%d", &n);
11
12    while (s <= n)
13    {
14        c++;
15        i++;
16        c++;
17        s += i;
18        c++;
19    }
20
21    c++;
22
23    printf("%d", c+2);
24
25    return 0;
26 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓